

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background.

A COMPREHENSIVE COVERAGE

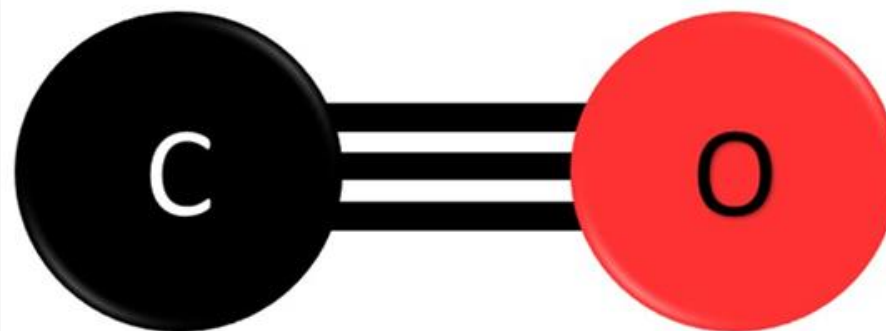
MODULE - 17

AIR POLLUTION - 4

MAJOR GASEOUS AIR POLLUTANTS - THEIR SOURCES & EFFECTS

1 CARBON MONOXIDE (CO)

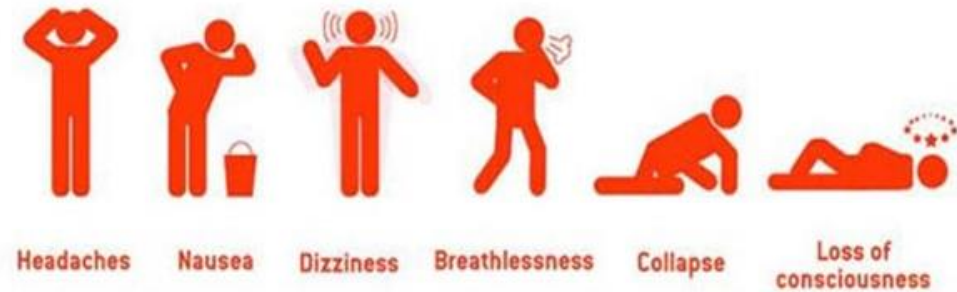
- Carbon monoxide (CO) is a colorless, odorless and highly toxic gas that is slightly less dense than air.
- Carbon monoxide is produced from the incomplete combustion of fossil fuels. It forms when there is not enough oxygen to produce carbon dioxide (CO₂).
- Metallurgical Industries (Iron smelting) also produce carbon monoxide as a by-product.
- The largest source of carbon monoxide is natural, due to photochemical reactions in the troposphere.
- The natural sources of CO are: photochemical reactions in the troposphere, volcanoes, forest fires, etc.



EFFECTS ON HEALTH

- Carbon monoxide poisoning is the most common type of fatal air poisoning.
- It combines with hemoglobin to produce Carboxyhemoglobin, which is incapable of carrying oxygen to different cells and tissues in the body.

Signs of carbon monoxide poisoning



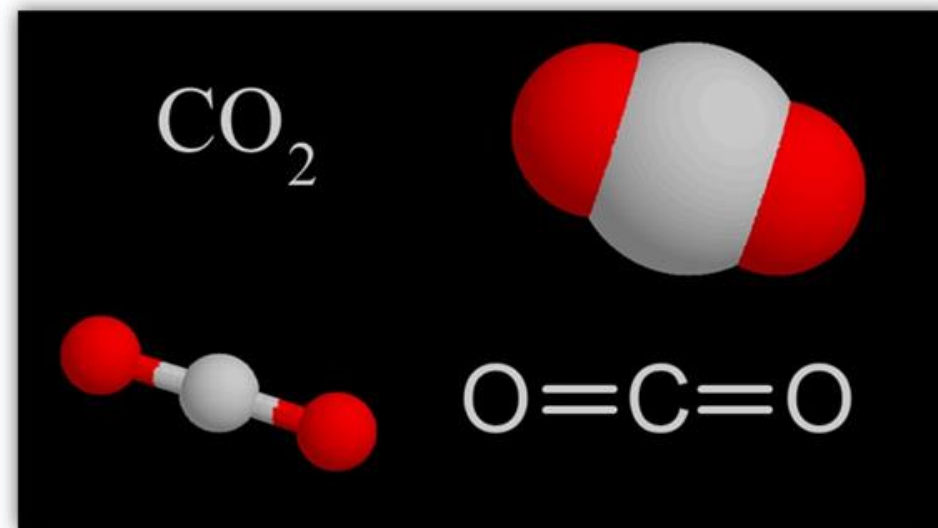
EFFECTS ON ENVIRONMENT

In the atmosphere, through photo-chemical reactions CO plays a role in the formation of ground-level ozone (tropospheric ozone) and can also increase the concentrations of methane.

MAJOR GASEOUS AIR POLLUTANTS - THEIR SOURCES & EFFECTS

2 CARBON DIOXIDE (CO₂)

- Colorless and odorless gas vital to life on Earth. It is heavier than air.
- Natural sources include volcanoes, hot springs and geysers, and it is released from carbonate rocks due to weathering.
- Because carbon dioxide is soluble in water, it occurs naturally in groundwater, rivers and lakes, in ice caps and glaciers and also in seawater.



EFFECTS ON HEALTH

CO₂ is an asphyxiant gas (asphyxia → a condition arising when the body is deprived of oxygen, causing unconsciousness or death.).



EFFECTS ON ENVIRONMENT

- Carbon dioxide is an important greenhouse gas.
- Burning of carbon-based fuels since the industrial revolution has led to global warming.
- It is also a major cause of ocean acidification because it dissolves in water to form carbonic acid.



MAJOR GASEOUS AIR POLLUTANTS - THEIR SOURCES & EFFECTS

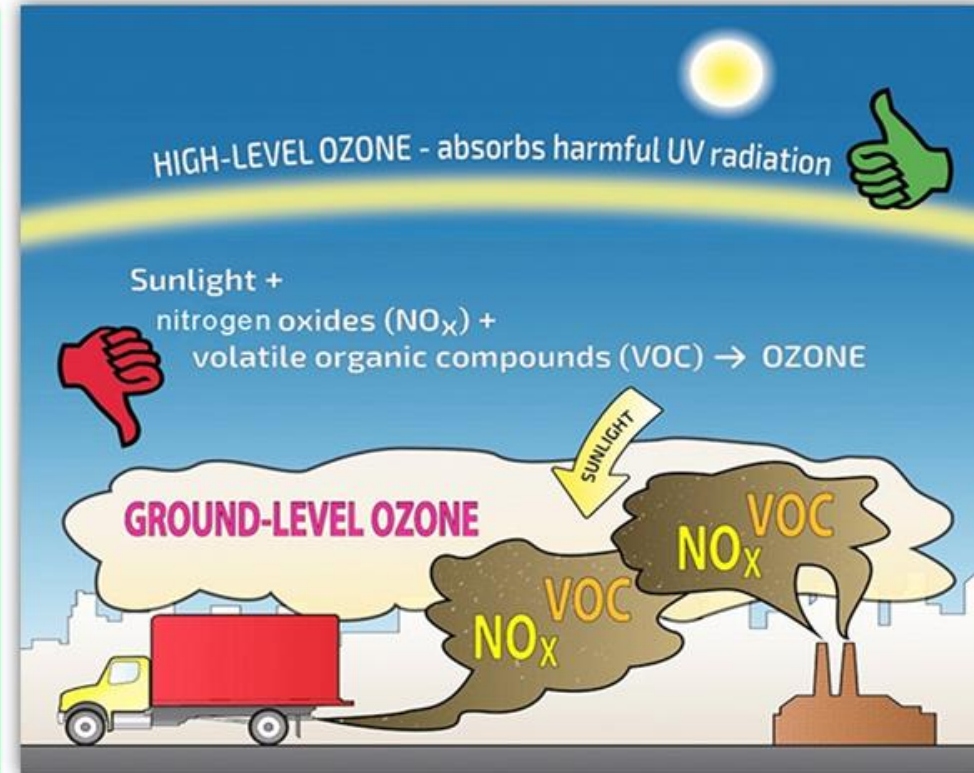
3 GROUND-LEVEL/TROPOSPHERIC OZONE (O_3)

- It occurs naturally in the Stratosphere in the form of ozone layer where it absorbs harmful ultraviolet rays of the sun and protects the life on the planet.
- It also occurs at Troposphere naturally but in low concentrations.
- At the ground level, it is produced due to human activities. It is a pollutant with toxic effects.
- Vehicles and industries are the major source of ground-level ozone emissions.
- Carbon monoxide, Nitrogen dioxide also play a major role in the formation of Ozone at the ground level through photo-chemical reactions.



EFFECTS ON HEALTH

- High concentrations of ozone near ground level can be harmful to people, animals and crops.
- Ozone irritates the eyes and respiratory system, causing cough and irritation in the throat.
- Ozone can aggravate asthma and other chronic lung diseases, and can inflame and damage cells in the lungs.
- Finally, ozone also causes permanent lung damage in the long run.



EFFECTS ON ENVIRONMENT

Ground level ozone is a powerful greenhouse gas that causes global warming.

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